

LANGMUIR'S SIX CASES: QUICK REFERENCE TABLE

Summary of All Adsorption Classifications (1918)

Case	Name	Key Feature	Equation	n-value	When to Use	Fluoride Relevance
I	Single-Site Langmuir (SSL)	1 molecule = 1 site	$= \frac{KP}{(1+KP)}$	$n = 1$	Homogeneous surface, low coverage	PRIMARY MODEL
II	Multi-Site Langmuir (MSL)	1 molecule = n sites	$= \frac{(KP)^{1/n}}{(1+(KP)^{1/n})^{1/n}}$	$n > 1$	Large molecules, polymers	If F aggregates
IIa	Dual-Site Langmuir (DSL)	2 site types (strong+weak)	$= + -$		Heterogeneous surface	LIKELY for coconut husk
III	Generalized Langmuir (GL)	Mobile molecules	Complex (partition function)	-	High T, weak binding	Not relevant
IV	Cooperative Adsorption (CA)	Neighbor interactions	$= \frac{(KP)}{(1+(KP))}$	$n = 1$	Self-assembly, proteins	pH-induced repulsion?
V	Dissociative Adsorption (DA)	Molecule splits ($A \rightarrow 2A$)	$= \frac{(KP)^{1/2}}{(1+(KP)^{1/2})^{1/2}}$	$n = 1/2$	H, O on metals	F doesn't dissociate
VI	Multilayer Adsorption (MLA)	Stacking layers	$= \frac{K P}{(1+K P+K K P^2+...)}$	-	High P, condensation	Low conc. range

Decision Tree: Which Langmuir Model to Use?

START: Fluoride on Coconut Husk

Is coverage < 70%?

YES → Try Case I (SSL) first

NO → Consider multilayer (unlikely at mg/L range)

Does residual plot show patterns?

NO → SSL is sufficient

YES → Check which pattern:

Bimodal → Try DSL (Case IIa)

pH-correlated → SSL won't work, need hybrid

Time-dependent → Kinetic model needed

Is $R^2 < 0.85$ for SSL?

YES → Surface is heterogeneous or pH matters

→ Use Hybrid (SSL + ML corrections)

NO → SSL is adequate, celebrate!

Parameter Extraction Guide

From Case I (SSL) Fit:

```
from scipy.optimize import curve_fit

def langmuir(Ce, qmax, KL):
    return (qmax * KL * Ce) / (1 + KL * Ce)

# Fit
params, cov = curve_fit(langmuir, Ce_data, qe_data)
qmax, KL = params

# Extract uncertainties
perr = np.sqrt(np.diag(cov))
qmax_err, KL_err = perr

# Calculate R²
from sklearn.metrics import r2_score
qe_pred = langmuir(Ce_data, qmax, KL)
r2 = r2_score(qe_data, qe_pred)

print(f"qmax = {qmax:.2f} ± {qmax_err:.2f} mg/g")
print(f"KL = {KL:.4f} ± {KL_err:.4f} L/mg")
print(f"R² = {r2:.4f}")
```

From Case IIa (DSL) Fit:

```
def dual_site_langmuir(Ce, qmax1, KL1, qmax2, KL2):
    site1 = (qmax1 * KL1 * Ce) / (1 + KL1 * Ce)
    site2 = (qmax2 * KL2 * Ce) / (1 + KL2 * Ce)
    return site1 + site2
```

```
# Fit with initial guesses
p0 = [qmax1_guess, KL1_guess, qmax2_guess, KL2_guess]
params, cov = curve_fit(dual_site_langmuir, Ce_data, qe_data, p0=p0)
qmax1, KL1, qmax2, KL2 = params

print(f"Strong sites: qmax1 = {qmax1:.2f} mg/g, KL1 = {KL1:.4f} L/mg")
print(f"Weak sites: qmax2 = {qmax2:.2f} mg/g, KL2 = {KL2:.4f} L/mg")
```

Interpretation Checklist

After fitting Langmuir, ask these questions:

Goodness of Fit

- ☐ $R^2 > 0.85$? (acceptable)
- ☐ $R^2 > 0.90$? (good)
- ☐ $R^2 > 0.95$? (excellent)

Physical Reasonableness

- ☐ qmax in literature range? (6-10 mg/g for coconut)
- ☐ KL in literature range? (0.08-0.15 L/mg)
- ☐ RL between 0 and 1? (favorable)

Residual Analysis

- ☐ Residuals normally distributed?
- ☐ No correlation with pH?
- ☐ No correlation with time?
- ☐ No correlation with temperature?

When to Use Dual-Site (DSL)

- ☐ Single SSL fit poor ($R^2 < 0.85$)?
 - ☐ Residuals show two populations?
 - ☐ Expect surface heterogeneity (edges + basal)?
 - ☐ DSL improves R^2 significantly (>0.05)?
-

Langmuir vs. Your Hybrid Model

Aspect	Pure Langmuir (Case I)	Hybrid (SSL + ML)
Physics	Full (equilibrium)	Full (baseline)
pH effects	Not modeled	ML learns
Kinetics	Assumes equilibrium	ML learns

Aspect	Pure Langmuir (Case I)	Hybrid (SSL + ML)
Temperature	Needs $K(T)$ fit	ML learns
Interpretability	Excellent	Good (decomposed)
Extrapolation	Risky	Risky (but constrained)
Data needed	Low (~20 points)	Medium (~50-80 points)
Complexity	Low (2 params)	Medium (2 + 8-12 features)
R^2 expected	0.85-0.90	0.90-0.95

Historical Timeline

1907 Freundlich isotherm (empirical)
 ↓
 1916 Langmuir begins adsorption studies
 ↓
 1918 LANGMUIR PUBLISHES 42-PAGE PAPER
 - 6 cases defined
 - Kinetic derivation
 - Experimental validation
 ↓
 1932 Langmuir wins Nobel Prize
 ↓
 1938 BET theory (multilayer extension)
 ↓
 1940s Temkin isotherm (heat of adsorption)
 ↓
 1960s Dubinin-Radushkevich (micropores)
 ↓
 1990s Sips/Langmuir-Freundlich (hybrid)
 ↓
 2000s Machine learning + physics hybrids
 ↓
 2024 YOUR PROJECT: SSL + ML residuals

Common Mistakes to Avoid

Mistake 1: Linearizing the Equation

Wrong:

DON'T DO THIS!

$y = C_e / q_e$

$x = C_e$

`slope, intercept = np.polyfit(x, y, 1)`

Right:

```
# DO THIS!
params, cov = curve_fit(langmuir, Ce, qe)
```

Why? Linear transforms distort error structure and give wrong parameter uncertainties.

Mistake 2: Mixing pH Values

Wrong:

```
# Fit all data together regardless of pH
params = curve_fit(langmuir, Ce_all, qe_all)
```

Right:

```
# Fit separately per pH or include pH in model
for pH_val in [3, 5, 7, 9]:
    data_at_pH = df[df['pH'] == pH_val]
    params = curve_fit(langmuir, data_at_pH['Ce'], data_at_pH['qe'])
    print(f"pH {pH_val}: qmax={params[0]:.2f}, KL={params[1]:.4f}")
```

Mistake 3: Over-Interpreting R^2

Wrong thinking: “ $R^2 = 0.92$, so Langmuir is perfect!”

Right thinking: “ $R^2 = 0.92$ means 92% of variance explained. Check residuals for systematic patterns that ML could learn.”

Mistake 4: Ignoring Uncertainty

Wrong:

```
print(f"qmax = {qmax}") # No error bars!
```

Right:

```
perr = np.sqrt(np.diag(cov))
print(f"qmax = {qmax:.2f} ± {perr[0]:.2f} mg/g")
```

End of Quick Reference Table

Use this table to decide which Langmuir variant to use and how to interpret results!